13/26 EXP NO.-08 TRANSIENT ANALYSIS OF SERIES RLC CIRCUIT

AIM:

To determine transient response of a Series RLC circuit using Multisim simulator.

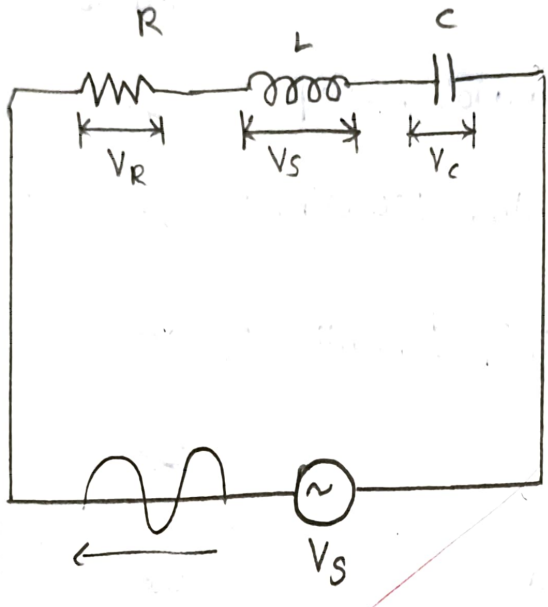
APPARATUS REQUIRED:

S.No.	COMPONENTS	SPECIFICATION	QUANTITY
1.	CRO	10MHz	1
2.	DCB, DIB	—	1
3.	Resistors	10kΩ	—
4.	Bread boards	—	1
5.	Connecting wires	—	—
6.	PC with multisim simulator	—	—

CIRCUIT DIAGRAM:

Choose $R = 10k\Omega$, $L = 10mH$ and $C = 0.1\mu F$

(values of R, L and C can be changed)



PROCEDURE :

1. Make connections as per the circuit diagram.
2. Calculate the values of $V_C(t)$, $V_R(t)$, $V_L(t)$ and write in table.
3. Compare the calculated values and measured values of voltage in both the cases.

TABULATION :

Voltage across capacitor :

Time (t)	$V_c(t)$ - charging cycle (volts) Observed values	$V_c(t)$ - Discharging cycle (volts) Observed values
$t = 0$	0	V
$t = \tau$	0.632V	0.368V
$t = 2\tau$	0.865V	0.135V
$t = 3\tau$	0.950V	0.050V
$t = 4\tau$	0.982V	0.018V
$t = 5\tau$	0.993V	0.007V

Voltage across inductor :

Vin amplitude : 10V , Vin frequency : 1KHz

time constant τ : $L/R = 2/R$

Time (t)	$V_L(t)$ = Voltage across inductor (observed values)
$t = 0$	10V
$t = \pi$	3.68V
$t = 2\pi$	1.35V
$t = 3\pi$	0.50V
$t = 4\pi$	0.18V
$t = 5\pi$	0.07V

O/P verified

$V_L(t) = V_{in} e^{-t/\tau} \Rightarrow t = 0 \Rightarrow V_L = 10e^{-0} = 10V$

Voltage acros resistor :

Time (t)	$V_R(t)$ - Voltage acros resistor Observed values
$t = 0$	0
$t = \tau$	6.32
$t = 2\tau$	8.65
$t = 3\tau$	9.50
$t = 4\tau$	9.82
$t = 5\tau$	9.93

$$V_2(t) = Ve^{-t/Re} = Ve^{-t/\tau}$$

Voltage acros capacitor :

$$V_c(t) = V(1 - e^{-t/\tau})$$

$$t = \tau$$

$$V_c = 1(1 - e^{-1}) = 1(1 - 0.368)$$

$$= 0.632V$$

$$t = 2\tau$$

$$V_c = 1(1 - e^{-2}) = 1(1 - 0.135)$$

$$= 0.865V$$

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S.no	Description	Marks alloted	Marks obtained
1.	Experimental / setup	20	20
2.	Observation / reading	20	20
3.	Calculation / result	20	20
4.	Viva	20	20
5.	Record	20	20
	Total =	100	100

100

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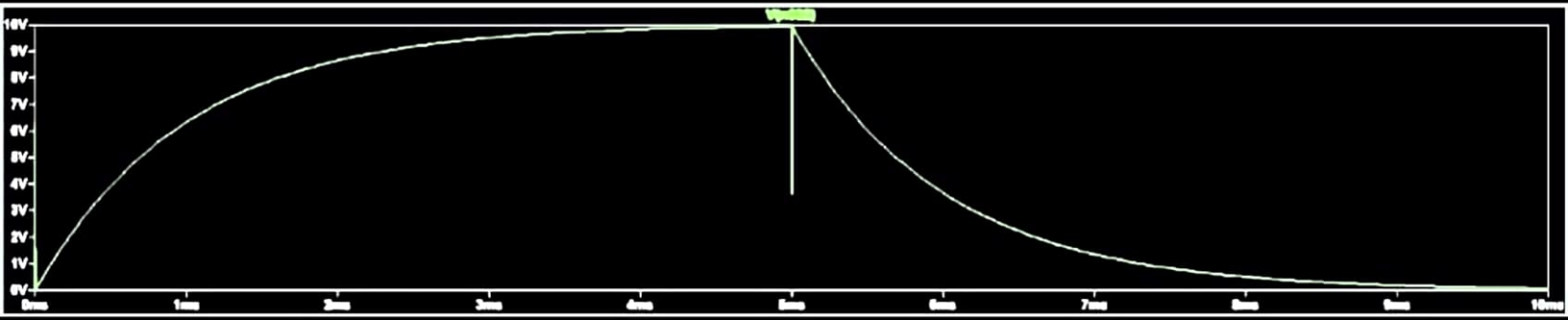
RESULT :

Thus, the transient analysis of series RLC circuit is done practically using Multisim Simulator.


10V
9V
8V
7V
6V
5V
4V
3V
2V
1V
0V
0ms
1ms
2ms
3ms
4ms
5ms
6ms
7ms
8ms
9ms
10ms